David Robert

905 375 4174

david.connor.r@gmail.com

Portfolio:

www.davidrobert.computer

Creative Technologist & Interactive Systems Designer

I apply emerging technologies to interactive experiences in the physical world.
Experienced across art studios, design agencies, and research labs.
My work has been shown internationally and engaged by thousands of participants in public space.
Focused on real-time interactive systems, generative & AI pipelines, physical computing, and dependable on-site delivery.

EXPERIENCE

Creative Technologist (R&D)

Antimodular Research Inc.,
Atelier Lozano-Hemmer
Montreal QC
November 2023 - September 2024
January 2025 - Present

- Built and deployed real-time interactive systems for public artworks by Rafael Lozano-Hemmer, turning conceptual proposals into durable interfaces in Python and TouchDesigner that have been used by tens of thousands of participants.
- Led development of multimodal AI systems spanning computer vision, audience tracking, semantic retrieval, generative text, generative audio, and spatialized sound for large-scale public installations. For Undercurrents, this included a real-time system that listens to visitors and responds with contextually selected audio.
- Integrated and networked installation hardware including RealSense depth cameras, LiDAR, IP and USB cameras with computer vision, ArtNet and DMX lighting at scale, projectors, and Dante audio. Linked multi-computer setups over OSC, MQTT, HTTP, and NDI, and chose media formats and codecs such as HAP for reliable playback.
- Installed and supervised exhibitions on-site internationally, including a month-long install in Mexico City for a three-month outdoor exhibition, a month supervising a show in Abu Dhabi, and a 15-day install of Undercurrents in Houston. Worked directly with curators, collectors, and museum teams, and with venues and outside contractors on power, safety, and fabrication.
- Contributed to project scoping, hardware procurement, timeline estimation, and technical planning, with autonomy over language choice and project architecture. Kept work in Git, wrote Windows startup, shutdown, and batch automation (including a PowerShell and ffmpeg pipeline that fed Shadow Tuner), and documented every artwork with guides and manuals for the developers who maintain them.

Media Artist in Residence

Fabrica Research Centre
Treviso TV, Italy
September 2024 - December 2024

- Selected for an international artist residency to conceive, develop, and exhibit an interactive installation exploring AI mimicry and vocal embodiment, built with TouchDesigner and Python.
- Developed the work through critique and cross-disciplinary exchange with an international cohort, centered on the theme of Ruralism.

EDUCATION

BA in Media Production
Concentration in Digital Media
4.27 / 4.33 GPA
Toronto Metropolitan University
Toronto ON
September 2019 – April 2023

PUBLICATIONS

David Bouchard, Cintia Cristia, David Robert, Michael Bergmann. 2022. Augmented Symphony: An augmented reality application for immersive music listening. In *Proceedings of EVA London 2022.* Computer Arts Society, London, England, UK.

EXHIBITIONS

David Robert. I SURRENDERED MY BODY AND I SUCCUMBED TO THE BEAST. At *Until the Very End.* Treviso, Italy. *Contribution: Artist.*

David Robert, Alex Verni, Design + Technology LAB. 2022. Assembly Line. At *DesignTO 2022.* Toronto, Canada. *Contribution: Lead Developer and Designer.*

David Robert, Alex Verni, Stacy Cernova, Anthony Baloukas. 2022. Moth Melody. At *Ontario Science Centre.* Toronto, Canada. *Contribution: Lead Developer.*

SKILLS

Programming

-Python
-JavaScript
-Node.js
-React
-GLSL

Interactive Media

-TouchDesigner
-Unreal Engine
-Unity
-Three.js
-ComfyUI

Systems & Networking

-OSC
-MQTT
-NDI
-WebSockets
-REST APIs
-Git

Physical Computing

-Arduino
-Raspberry Pi
-DMX / ArtNet
-RealSense
-Circuit design
-3D printing
-Laser cutting

Creative Technologist / Lab Assistant
Design + Technology LAB,
Toronto Metropolitan University
Toronto ON
September 2021 - April 2022
September 2022 - April 2023
September 2023 - November 2023

- Assisted in day-to-day operations of the lab, including maintenance and operation of lab equipment. This includes 3D printing, laser cutting, XR, and robotics.
- Created an interactive 3D projection-mapping installation for DesignTO 2022 using TouchDesigner, OpenVR, a KUKA robot arm, and full-stack web development.

Research Assistant [AI]
Technology Research in Performance Lab,
Toronto Metropolitan University
Toronto ON
September 2022 - April 2023

- Researched human–AI co-performance, building experimental systems that merged live theatre with generative AI.
- Built a real-time theatrical AI-improvisation system using cloud-based AI tools, serving as an early prototype of interactive dialogue with AI in public performance.

Teaching Assistant [Physical Computing]
Toronto Metropolitan University
Toronto ON
September 2022 - December 2022

- Assessed students and gave feedback on projects for RTA 321: Intro to Tangible Media.
- Supported students with Arduino programming and simple circuit design.

Creative Developer Intern
Jam3
Toronto ON
May 2022 - August 2022

- Built a multi-user online 3D space in Three.js with persistent data (React, Node, MongoDB, WebSockets), showing how small businesses could interact with the Metaverse. Delivered as a client-facing prototype.
- Worked in a fast-paced, autonomous intern team of developers, designers, and a coordinator.

Research Assistant [AR]
Toronto Metropolitan University
Toronto ON
July 2021 - April 2022

- Developed an AR app using Unity [C#] centred on enhancing digital delivery of orchestral music through spatial and interactive audio.
- Co-authored peer-reviewed paper accepted to EVA London 2022.

ADDITIONAL EXPERIENCE

Undergraduate Researcher
[Sim Developer & HRI Designer]
Synaesthetic Media Lab,
Toronto Metropolitan University
Toronto ON
September 2022 - December 2022

- Collaborated with a joint university research team to prototype interactive, networked “boat-drones” for live performance environments.
- Designed and programmed a simulation environment in Unreal Engine 5 (C++) and built real-time communication systems (Python server, WebSockets, I2C) to synchronize physical and virtual performance elements.

Interactive Developer
Ontario Science Centre
Toronto ON
January 2022 - April 2022

- Designed and delivered a projection-based audio installation that engaged families and children, turning abstract tech into playful, intuitive interaction. Ensured reliability for constant public use.

Undergraduate Researcher
[Unreal AR Developer]
Synaesthetic Media Lab,
Toronto Metropolitan University
Toronto ON
January 2021 - April 2021

- Designed and developed an AR HoloLens 2 application in Unreal Engine 4 [Blueprints].
- The app supports training for medical workers through the manipulation of medical models with gestures and tools.

Fabrication Technician
Philip Beesley Studio,
Living Architecture Systems Group
Toronto ON
February 2020 - March 2020

- Fabricated components for an interactive installation that was part of the 2021 Venice Biennale of Architecture.